Rupak Tiwari

Kathmandu, Nepal · +9779864830964 ·
rupaktiwari18@gmail.com
Google Scholar · Hugging Face · GitHub · LinkedIn

AI Engineer and Researcher building LLM, RAG and ML systems, with a research background in low-resource speech recognition and machine translation.

EXPERIENCE

Information and Language Processing Research Lab (ILPRL) Dhulikhel, Nepal

Junior Research Assistant (JRA) and Research Collaborator *Aug 2025 – Sept 2026*

- Developed a Tamang-Nepali-English Tri-Lingual Machine Translation (TMT) system, funded by Google Academic Research under the Society Centered AI category.
- Fine-tuned open-source MT models for bilingual and trilingual purpose and evaluated them on standard metrics.
- Worked on corpus collection and fine-tuning for Newari Automatic Speech Recognition (ASR).
- Submitted a manuscript of ShrutiParichhya: A Nepali ASR Evaluation Dataset and Error Analysis of Fine-Tuned ASR Models to IJCA, performing acoustic and linguistic error analysis of various Nepali ASR models.

Research Intern *Mar 2025 – July 2025*

- Fine-tuned Wav2Vec2, Whisper and MMS models for Nepali ASR ,achieving highly competitive Word Error Rates (WER).
- Manually collected and curated 5 hours of Nepali speech corpus for Nepali ASR evaluation.

SKILLS

- **AI & Machine Learning:** Python, PyTorch, Hugging Face Transformers, Model Fine-tuning, RAG, LangChain, LangGraph, Scikit-learn, Pandas, NumPy, Weights & Biases, LLM Evaluation,
- **Backend & Infrastructure:** FastAPI, Pydantic, REST APIs, Docker, PostgreSQL, Git, Linux, Shell Scripting, MCP
- **AI Development Tools**: Claude Code

SELECTED PROJECT

Shruti — Nepali Speech Recognition & Conversational AI Assistant

- Fine-tuned Wav2Vec2-BERT on OSLR 54 Nepali speech data, tracking convergence with Weights & Biases over 136 GPU hours.
- Engineered a decoupled, quantized (FP32→INT8) inference pipeline with VAD for silence/noise filtering and reliable transcription.
- Designed intent routing across RAG, general responses, and human escalation, with ChromaDB using sentence-transformer embeddings and a custom KYC verification knowledge base.
- Implemented SQLite-backed persistent conversation state and logged structured escalation data.
- Exposed ASR and conversational workflows via FastAPI with a Streamlit frontend.
- Built a CI/CD pipeline with GitHub Actions to automate Docker image builds and deployment to Hugging Face Spaces.
- Containerized and deployed the backend using Docker, `uv`, Docker Hub, and Hugging Face Spaces.
- Prototyped real-time streaming ASR using WebSockets, VADIterator, and streamlit-webrtc, identifying quality challenges relative to batched ASR.

PUBLICATIONS

- **NepTam**: A Nepali-Tamang Parallel Corpus and Baseline Machine Translation Experiments *LREC 2026*
  Evaluated Nepali–Tamang parallel corpora and benchmarked multilingual MT models, primarily M2M-100.
- **Nwāchā Munā**: A Devanagari Speech Corpus and Proximal Transfer Benchmark for Nepal Bhasha ASR *CHiPSAL 2026*
  Contributed to the creation of a 5.39-hour transcribed Nepal Bhasha speech corpus and benchmarked cross-lingual transfer learning.
- **NepConformer**: A Conformer-Based Nepali Automatic Speech Recognition System *CML 2025/Springer*
  Worked on end-to-end Conformer ASR model on the OSLR54 dataset, achieving state-of-the-art Nepali ASR performance with Character Error Rate (CER) of 6.01%.

EDUCATION

Kathmandu University Dhulikhel, Nepal

Bachelor of Science in Computer Science (CGPA: 3.35/4.00) *Feb 2021 – Sept 2025*